

Causes of the instability: There are two main sources of instability. First, there could be an incompatibility between the AGP card and the slot on the motherboard. Second, AGP instability will also be noticed on cards and motherboards configured for 8x speeds. Sometimes, at this speed, the data transfer rate of 2133 MBps becomes too fast for the rest of the system to handle.

The motherboard is a complex knit of multiple components, talking to each other at varying speeds. When the CPU, RAM, PSU and other components are all perfectly matched and adjusted, AGP should work just fine. Any weakness in any of the links in this chain and problems are bound to show up.

If you are facing reliability and stability problems with your system and you have narrowed it down to the AGP port, the solution is to verify card and motherboard compatibility and/or slow down AGP and see if that solves the problems. In rare cases, you might also face a memory resource conflict.

Solutions

Incompatibility: The AGP card communicates with the AGP motherboard using a 'signalling voltage' that varies according to the version. AGP 1.0 uses 3.3 volts, AGP 2.0 uses 1.5 volts, and AGP 3.0 uses 0.8 volts. Depending on the AGP version supported, the AGP card has one or two slots. 3.3 volts uses one type of slot, while 1.5 and 0.8 volts use the same type of slot. According to the official AGP 3.0 specifications, AGP cards have been categorised as in the table on the next page (*AGP Card Specifications*).

The above are Intel's official names for the cards allowed by the specification. Manufacturers, however, are not so consistent, and you might find various kinds of names describing their cards. So use the table on the next page to figure out what kind of AGP card you have. Similarly, AGP slots on the motherboard also have official specifications as in the second table (*AGP Slot Specifications*).